

Accessibility Audit Brief: mariahllopez.com (portfolio)

Context

The website evaluated is a personal portfolio site, *mariahllopez.com*, designed to showcase skills, projects, and professional experience in analytics engineering, digital product design, and UX. The intended audience includes recruiters, hiring managers, and collaborators in technical, UX, and data-focused roles.

Evaluation Method

The site was evaluated using the WAVE accessibility tool alongside a manual review. WAVE was used to identify errors, contrast issues, structural elements, and ARIA usage. Manual testing included reviewing heading hierarchy, link clarity, and basic keyboard navigation. This combined approach ensured both automated detection and usability-focused evaluation.

Key Accessibility Barriers

1. Low Contrast Text (14 Errors)

- **Issue:** WAVE identified 14 contrast errors, indicating text elements with insufficient color contrast.
- **Evidence:** Contrast ratio issues were flagged across multiple UI elements despite some combinations passing (e.g., blue on white).
- **WCAG Guideline: WCAG 2.1 – 1.4.3 Contrast (Minimum)**
- **Impact:** Users with low vision may struggle to read text, reducing usability and accessibility.
- **Recommendation:**
Adjust color combinations to meet at least a **4.5:1 contrast ratio** for normal text. Use darker text colors or lighter backgrounds consistently across all components.

2. Suspicious Alternative Text (3 Alerts)

- **Issue:** Some images contain unclear or potentially non-descriptive alternative text.
- **Evidence:** WAVE flagged “suspicious alt text” warnings.
- **WCAG Guideline: WCAG 1.1.1 Non-text Content**
- **Impact:** Screen reader users may not understand the purpose or content of images.
- **Recommendation:**
Replace vague alt text with meaningful descriptions. For example:
 - Instead of: "image1"
 - Use: "Portrait of Mariah Lopez on homepage"

3. Redundant Links (2 Alerts)

- **Issue:** Multiple links with the same text point to the same destination.
- **Evidence:** Navigation and repeated “View Case Study” links were flagged.
- **WCAG Guideline: WCAG 2.4.4 Link Purpose (In Context)**

- **Impact:** Screen reader users may experience confusion or unnecessary repetition.
- **Recommendation:**
Make link text more specific, such as:
 - “View Case Study: Data Dashboard Project”
 - “View Case Study: UX Redesign”

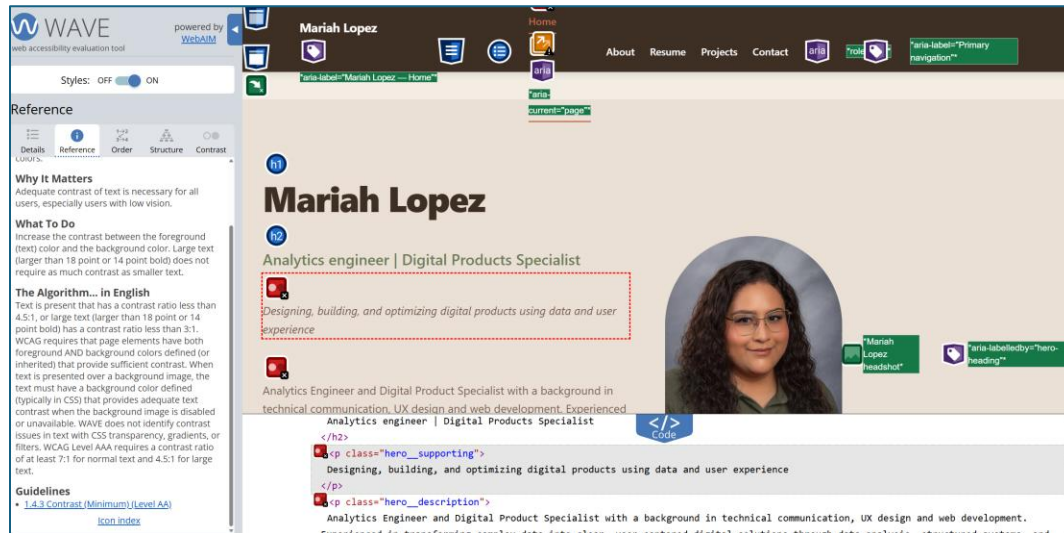


Figure 1: WAVE accessibility tool highlighting low contrast text on the homepage, indicating readability issues for users

4. Link to PDF Without Warning (1 Alert)

- **Issue:** A resume download link leads to a PDF without indicating file type.
- **WCAG Guideline: WCAG 2.4.4 Link Purpose**
- **Impact:** Users may not expect a file download or different format.
- **Recommendation:**
Update link text to:
 - “Download Resume (PDF)”

5. Heading Structure and Hierarchy

- **Issue:** The page includes multiple **H3 headings (9 total)** under H2 sections, which is mostly correct, but requires consistency checks.
- **Evidence:** WAVE structure panel shows:
 - 1 H1
 - 3 H2
 - 9 H3
- **WCAG Guideline: WCAG 1.3.1 Info and Relationships**
- **Impact:** Inconsistent or overly dense heading structures can reduce navigation clarity for screen readers.
- **Recommendation:**
Ensure headings follow a logical hierarchy and avoid skipping levels. Group related content clearly under each section.

6. ARIA Usage (22 Instances)

- **Issue:** ARIA labels and roles are present but may be overused or inconsistently applied.
- **Evidence:** WAVE identified 22 ARIA attributes.
- **WCAG Guideline: WCAG 4.1.2 Name, Role, Value**
- **Impact:** Incorrect or excessive ARIA can confuse assistive technologies.
- **Recommendation:**
Use semantic HTML first (e.g., <nav>, <main>) and only apply ARIA when necessary.
Validate ARIA roles for correctness.

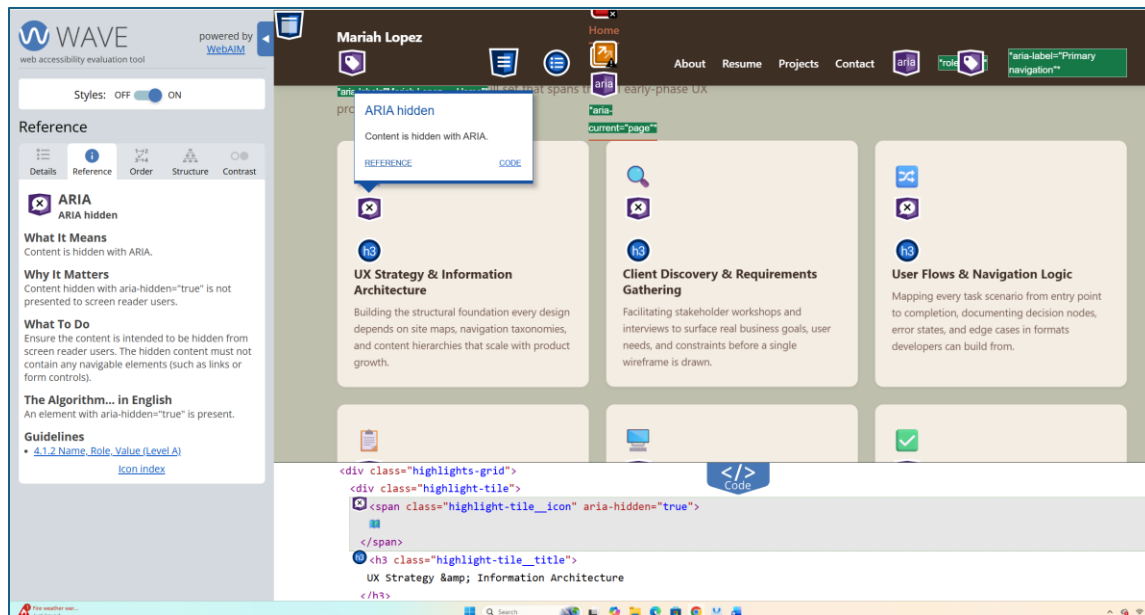


Figure 2: WAVE tool identifying ARIA-hidden content within feature cards, which may prevent screen readers from accessing important information.

Tool Limitations

While WAVE effectively identifies structural and contrast-related issues, it cannot fully evaluate usability or user experience. For example, it does not assess how intuitive navigation feels, how meaningful content is, or how well the site works with keyboard-only navigation. Manual testing is necessary to identify these deeper accessibility concerns.

Overall Assessment

The site demonstrates a strong structural foundation, including proper use of headings, navigation elements, and accessibility features such as skip links. However, improvements are needed in contrast, link clarity, and alternative text to meet WCAG standards more consistently. Addressing these issues will significantly enhance usability for users with disabilities and improve the overall accessibility score.

Accessibility Remediation Memo (UDOIT)

Purpose

The purpose of this remediation effort was to improve the accessibility of content within a Canvas sandbox course using the Universal Design Online Content Inspection Tool (UDOIT). UDOIT identifies accessibility issues and allows users to directly fix them within the course environment. This process focuses on applying accessibility best practices to ensure content is usable for all learners, including those using assistive technologies.

Summary of Accessibility Improvements

During the remediation process, multiple accessibility issues were identified and corrected using UDOIT's scanning and UFixIt tools. Key improvements included:

- Revising **non-descriptive links** to include meaningful, contextual text
- Addressing **image-based PDF files** by identifying the need for proper tagging and accessible formats
- Reviewing and improving **document accessibility options**, including alternate formats
- Ensuring **content structure and organization** followed accessibility best practices
- Resolving flagged issues across high, medium, and low impact categories

These changes improved the overall usability and accessibility of the course content.

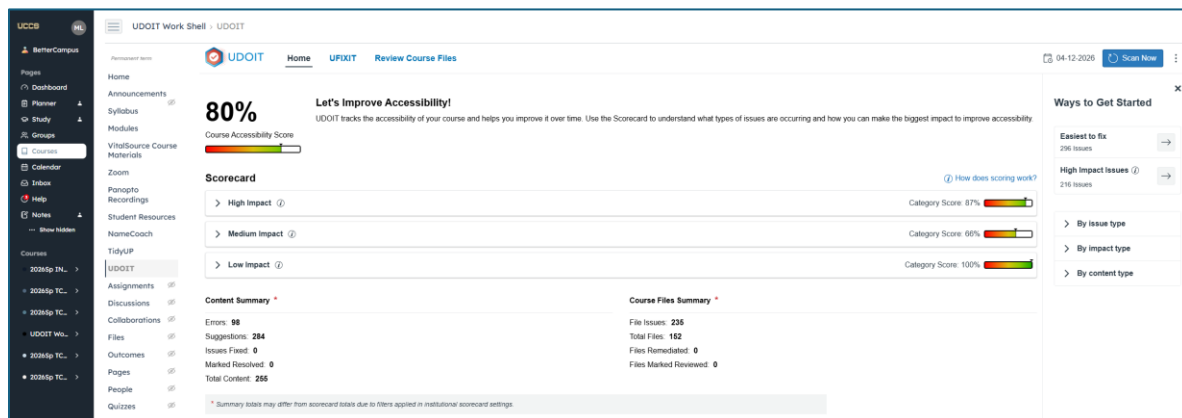


Figure 3: UDOIT scan results before remediation showing an 80% accessibility score with multiple errors and suggestions identified.

Evidence of Changes

Before Remediation

- Initial UDOIT scan showed a **Course Accessibility Score of 80%**
- The system identified:
 - **98 errors**
 - **284 suggestions**
 - **265 total content items**
- Issues included:

- Non-descriptive links , Image-based PDF files without proper tagging, and General content accessibility concerns across multiple categories

After Remediation

- After implementing fixes, the Course Accessibility Score improved to **100%**
- UDOIT results showed:
 - **0 issues across all high and medium impact categories**
 - All categories (color, images, links, headings) reached **100% compliance**
- This confirms that accessibility barriers were successfully addressed within the course environment.

Examples of Accessibility Fixes

1. Link Clarity Improvement

- Issue: Link contained a raw URL with no descriptive text
- Fix: Updated link text to clearly describe the destination
- Result: Improved readability and screen reader usability

2. PDF Accessibility Issue

- Issue: “Image-based file detected” (CORE-Phoneme-Deletion.pdf)
- Fix: Identified need for:
 - Text-based PDF conversion and Proper tagging structure
- Result: Improved compatibility with screen readers and assistive tools

3. Use of UFixIt Tool

- UFixIt allowed inline editing of accessibility issues
- Provided previews of changes before applying them
- Streamlined the remediation process directly within Canvas

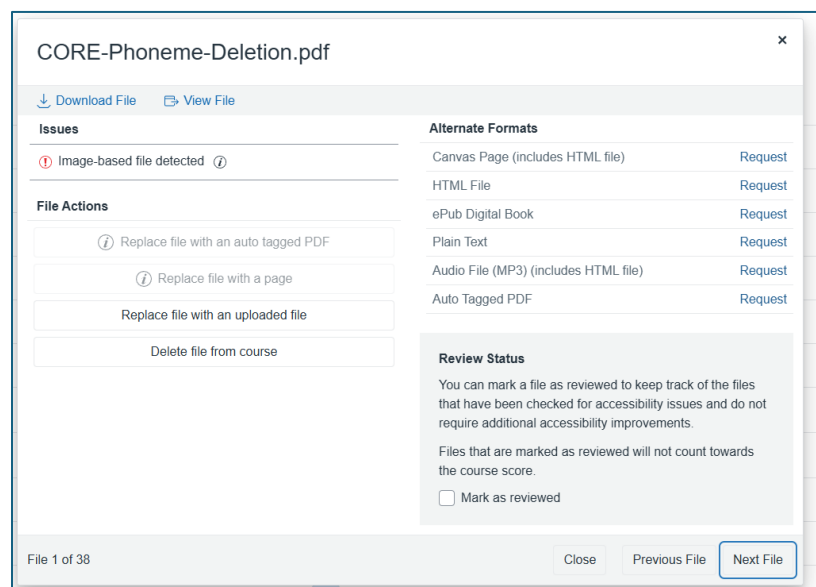


Figure 4: UDOIT identifying an image-based PDF file lacking accessibility features such as text recognition and tagging.

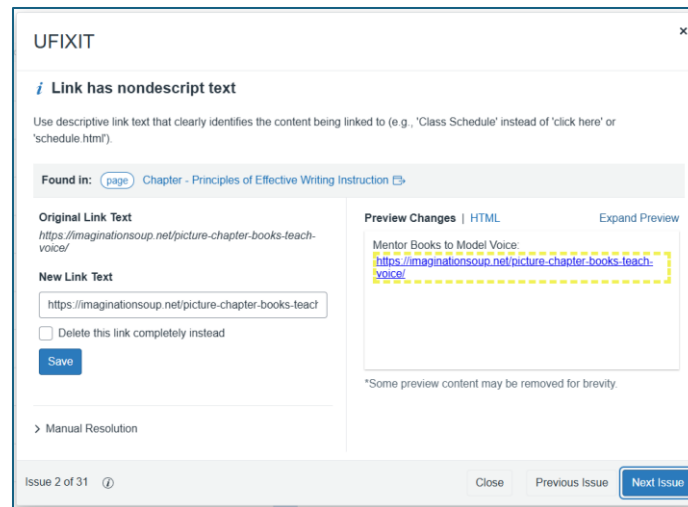


Figure 5: Demonstrating correction of a non-descriptive link by replacing a raw URL with meaningful link text.

Evidence of PDF Tagging

UDOIT identified documents that required accessibility improvements, particularly PDFs that were image-based. While full tagging requires external tools (such as Adobe Acrobat), UDOIT provided:

- Identification of inaccessible documents, Options to request **auto-tagged PDFs**, and Alternative formats such as:
 - HTML, ePub, Plain text , and Audio versions

This ensures content can be made accessible in multiple formats.

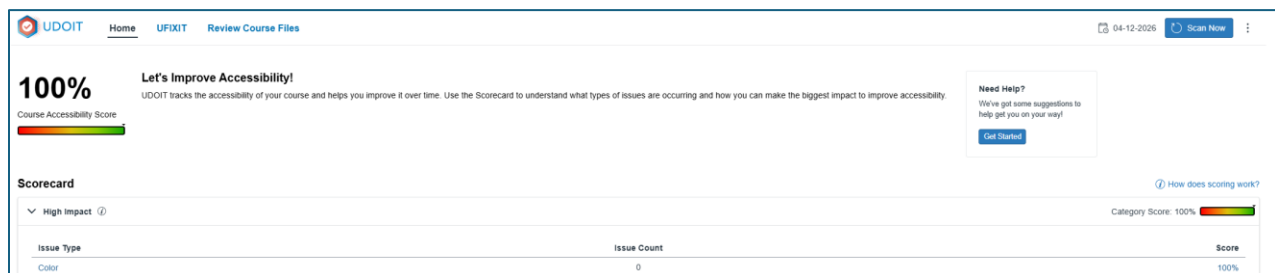


Figure 6: UDOIT scan results after remediation showing a 100% accessibility score with all issues resolved.

Lessons Learned

One key takeaway from this remediation process is that accessibility requires both automated tools and human decision-making. While UDOIT effectively identifies issues and provides guided fixes, it cannot fully evaluate context, clarity, or user experience. A challenge encountered was understanding how to properly remediate document-based issues, such as PDFs, which require additional tools outside of Canvas. Additionally, improving link clarity requires thoughtful rewriting rather than simple automated fixes. Overall, this experience reinforced the importance of integrating accessibility into content creation from the beginning. It also demonstrated how tools like UDOIT can significantly improve efficiency while still requiring critical thinking and manual review.